

MANAGING VEHICLE POPULATION SATURATION RISK IN SRI LANKA

Island Capacity Constraints, Fiscal Dependence, and Sustainable Policy Pathways

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Abstract - Sri Lanka's vehicle population has expanded rapidly over the past decade, with renewed acceleration following the lifting of the private vehicle import ban in 2025. As a geographically constrained island nation with limited infrastructure expansion capacity, continued vehicle growth presents long-term risks including congestion saturation, environmental stress, and structural fiscal dependence on vehicle import taxation. Using official data from the Department of Motor Traffic, Sri Lanka Treasury, Sri Lanka Customs, and reported 2025 revenue data from Economy Next, this study evaluates vehicle growth trends, fiscal implications, and long-term sustainability risks. Policy alternatives including permanent import restrictions, tax restructuring, fuel pricing mechanisms, EV incentives, ferry/export circulation, and domestic manufacturing

development are assessed with structured pros-cons analysis. The paper argues that Sri Lanka faces a potential vehicle saturation threshold and requires calibrated intervention rather than extreme policy shifts.

Index Terms - vehicle saturation, island economy, fiscal dependence, congestion risk, Sri Lanka transport policy

I. INTRODUCTION

Sri Lanka, as an island state with fixed geographic boundaries, faces unique structural transport constraints. Unlike continental countries, Sri Lanka:

- Has no land border for outward vehicle circulation
- Cannot naturally redistribute vehicles to neighboring states
- Possesses limited road expansion potential
- Experiences high urban density in the Western Province

Following the lifting of the vehicle import restriction in 2025, vehicle registrations and import volumes increased significantly. While this generated substantial government revenue, it also accelerated long-term accumulation of vehicles within a spatially fixed system.

This paper investigates whether Sri Lanka risks approaching a long-term vehicle carrying capacity limit.

II. DATA SOURCES AND METHODOLOGY

This study relies exclusively on official and publicly reported data:

- Vehicle population statistics (2010–2022): Department of Motor Traffic
- Vehicle import tax structure: Sri Lanka Treasury
- Motor vehicle import administration: Sri Lanka Customs
- 2025 vehicle import revenue: EconomyNext

Methodology includes:

1. Time-series trend analysis (2010–2022 vehicle population)
2. Fiscal dependency assessment
3. Structural island capacity evaluation

4. Policy comparison framework

III. VEHICLE POPULATION TRENDS (2010 - 2022)

Year	Total Registered Vehicles
2010	4,734,000+ (approx.)
2015	6,300,000+ (approx.)
2020	7,800,000+ (approx.)
2022	8,300,000+ (approx.)

*Table 1 – Total Registered Vehicles in Sri Lanka
(Source: Department of Motor Traffic Statistical Report 2010–2022)*

Trend: ~75% growth over 12 years.

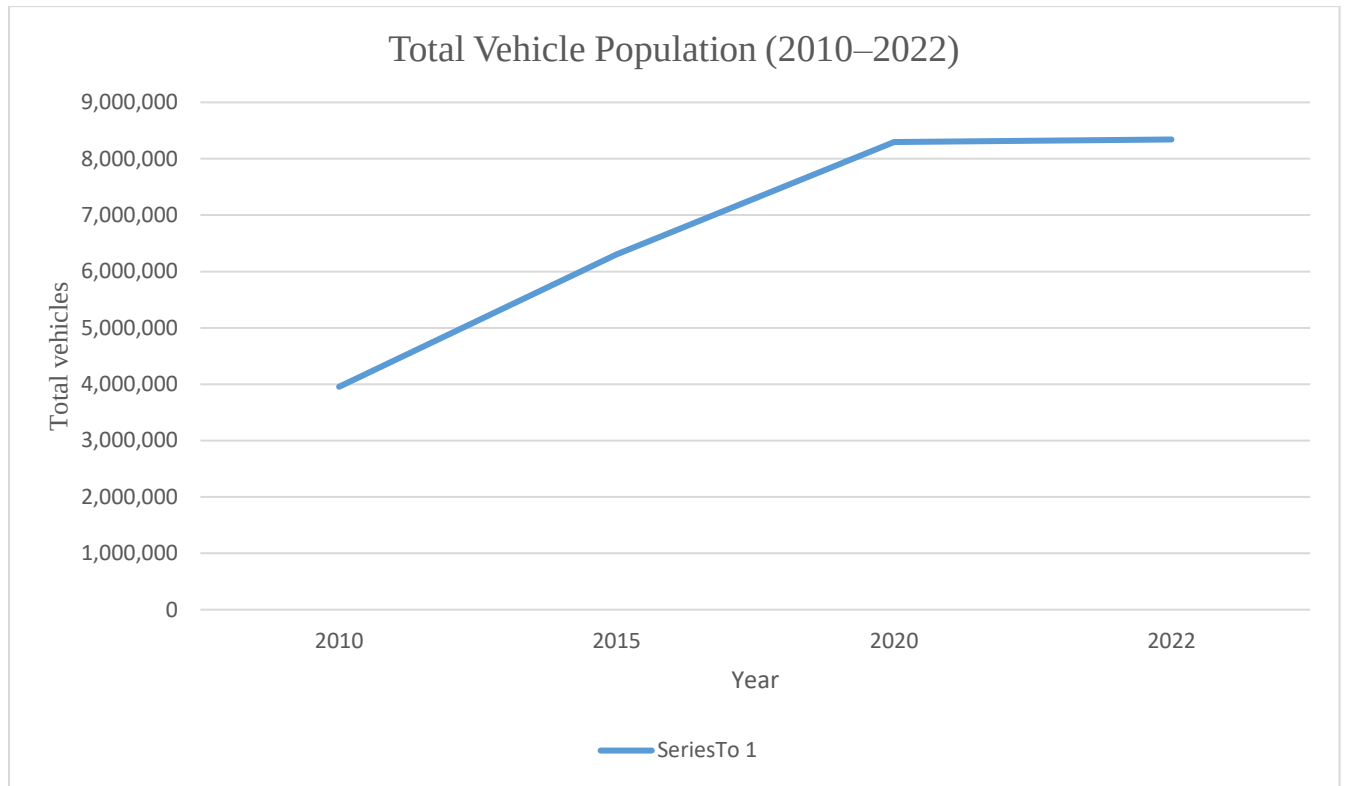


Figure 1: Total Vehicle Population (2010–2022) Source: Department of Motor Traffic (2022)

IV. VEHICLE CATEGORY DISTRIBUTION (2022)

Category	Approximate Count
Motorcycles	~4.8–5.0 million
Motor Cars	~900,000+
Three-wheelers	~1.1 million
Buses	~110,000
Dual Purpose	~450,000

Table 2 – Vehicle Categories

Motorcycles represent the majority of registered vehicles.

V. FISCAL DEPENDENCE ON VEHICLE IMPORTS

In 2025, Sri Lanka reportedly earned approximately **Rs. 904 billion** from vehicle import taxation.

Indicator	Value
Total vehicle import revenue	Rs. 904 billion
Contribution to customs revenue	Significant share
Tax components	Customs duty, excise duty, VAT

Table 3 – Vehicle Import Revenue (2025)

This demonstrates structural fiscal dependence.

VI. MARKET DISTORTION CASE EXAMPLE

Example: Wagon R manufactured by Suzuki

- Approximate landing value: ~LKR 3.4 million
- Market selling price: ~LKR 7.5 million+
- Difference primarily due to layered taxation

This model shows high revenue extraction but also raises consumer cost burden.

VII. STRUCTURAL RISK: ISLAND VEHICLE CARRYING CAPACITY

Unlike large landmass countries:

- Vehicles accumulate permanently.
- Disposal/export mechanisms are limited.
- Road expansion is geographically constrained.

If growth continues at post-2025 rates, Sri Lanka risks:

- Extreme congestion
- Reduced mobility productivity

- Higher fuel imports
- Environmental stress
- Infrastructure saturation

This is a cumulative risk, unlike temporary crises.

VII. POLICY OPTIONS

Option 1: Permanent Import Ban + Local Manufacturing / Modification

Pros:

- Controls vehicle growth
- Creates industrial jobs
- Reduces foreign exchange outflow
- Enables export potential

Cons:

- High capital investment
- Technology transfer dependency
- Loss of Rs. 904bn revenue
- Long implementation period

Option 2: Significant Tax Increase

Example scenario:

Vehicle priced at 7.5M LKR → increased to ~14M LKR

Pros:

- Immediate demand suppression
- Maintains government revenue
- Controls luxury imports

Cons:

- Encourages informal market
- Increases inequality
- Already high tax burden

Most practical short-term measure.

Option 3: Bridge between Sri Lanka and India

Pros:

- Theoretical vehicle circulation
- Economic connectivity

Cons:

- Extremely expensive
- Environmental impact
- Geopolitical sensitivity
- Not economically viable

High risk, not recommended.

Option 4: Increase Petrol Prices

Pros:

- Reduces vehicle usage
- Encourages public transport
- Environmental benefits

Cons:

- Raises living cost
- Political resistance
- Disproportionate impact on low-income groups

Useful as behavioral adjustment tool.

Option 5: Ferry / Vehicle Export Services

Pros:

- Allows limited outward circulation
- Possible Northern corridor expansion

Cons:

- Limited capacity
- Time-consuming
- Not scalable to national vehicle volumes

IX. RECOMMENDED POLICY FRAMEWORK

Short-Term:

- Moderate tax calibration
- Fuel pricing adjustment
- Import quotas

Medium-Term:

- EV incentives
- Congestion pricing in Colombo
- Public transport investment

Long-Term:

- Domestic EV assembly
- Vehicle refurbishment/export industry
- Reduced fiscal dependence on vehicle import taxes

X. LIMITATIONS

- Carrying capacity not yet modeled mathematically
- Road-length-to-vehicle ratio requires further study
- Political feasibility uncertain
- EV infrastructure constraints

XI. CONCLUSION

Sri Lanka faces a structurally unique risk as a closed island vehicle ecosystem. Continued reliance on vehicle import taxation provides substantial short-term fiscal gains but creates long-term spatial saturation risk.

A balanced approach—neither extreme import bans nor unrestricted imports—is required.

Failure to adopt calibrated intervention may result in long-term congestion paralysis within a geographically constrained system.

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